Quick tutorial for linear Gaussian state-space analysis of temperature time series with ThermoSST

Akira Hirao

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Appendix: Handling Raw Temperature Data 12

This appendix describes how the original temperature time series data are prepared prior to the state-space analysis presented in the main text. We outline the procedures for handling raw observations. 12

Set Ennvironment

```r
## Set libraries
library(ThermoSSM)
```

```r
library(tidyverse)
library(forecast)
library(cowplot)

# initialize
rm(list=ls(all=TRUE))
```

Loading Example Data

```r
# loading example data 1: daily mean sea surface temperature time series off southern Ibaraki Prefectur
data(ibaraki_sst)
head(ibaraki_sst)
```

```
##             Temp
## 1982-01-01 16.21
## 1982-01-02 16.28
## 1982-01-03 16.36
## 1982-01-04 16.36
## 1982-01-05 16.22
## 1982-01-06 16.04
```

Checking Example Data

```r
class(ibaraki_sst) # zoo object
```

```
## [1] "zoo"
```

```r
frequency(ibaraki_sst)   # frequency 1
```

```
## [1] 1
```

```r
start(ibaraki_sst)       # start time "1982-01-01"
```

```
## [1] "1982-01-01"
```

```r
end(ibaraki_sst)         # end time "2025-12-31"
```

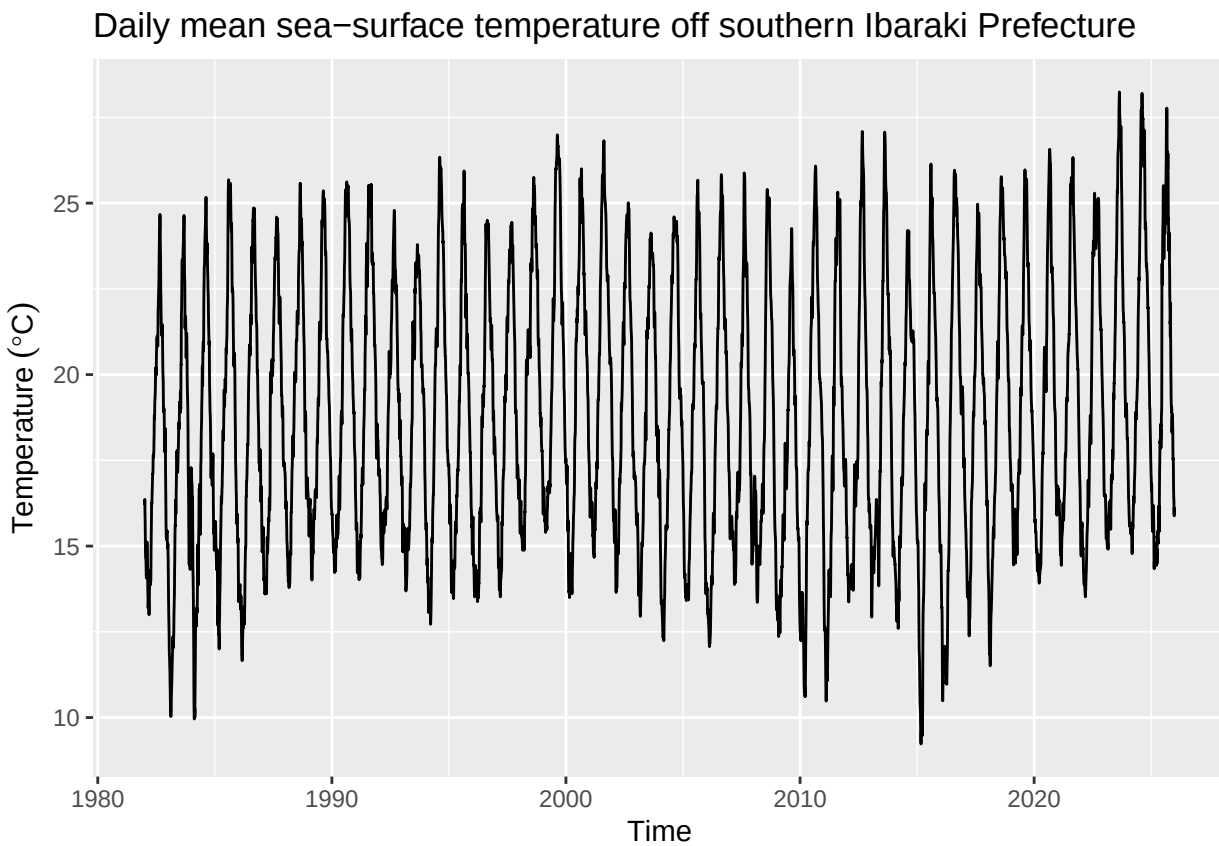
```
## [1] "2025-12-31"
```

Plotting Daily Time Series of Sea Surface Temperature

```r
daily_ibaraki_sst_plot <- forecast::autoplot(ibaraki_sst) +
  labs(y = expression(Temperature~(degree*C)),
       x = "Time") +
  ggtitle("Daily mean sea-surface temperature off southern Ibaraki Prefecture")

#ggsave("Daily_SST_ibaraki_plot.png",
#       plot=daily_ibaraki_sst_plot,
#       width=8,height=6)

plot(daily_ibaraki_sst_plot)
```



Converting from Daily zoo bject to Monthly ts object

```
monthly_ibaraki_sst <- zoo_daily2ts_monthly(ibaraki_sst)
```

Check of ts object

```
class(monthly_ibaraki_sst) # ts object
```

```
## [1] "ts"
```

```
frequency(monthly_ibaraki_sst)   # frequency 12
```

```
## [1] 12
```

```
start(monthly_ibaraki_sst)       # start time 1982, 1
```

```
## [1] 1982    1
```

```
end(monthly_ibaraki_sst)         # end time "2025, 12
```

```
## [1] 2025   12
```

```
cycle(monthly_ibaraki_sst) %>% head() #     1 12
```

```
##      Jan Feb Mar Apr May Jun
## 1982   1   2   3   4   5   6
```

```r
time(monthly_ibaraki_sst)  %>% head() #   1982.000, 1982.083...
```

```
##           Jan      Feb      Mar      Apr      May      Jun
## 1982 1982.000 1982.083 1982.167 1982.250 1982.333 1982.417
```

```r
window(monthly_ibaraki_sst, start = c(2001, 1), end = c(2001, 12))  #
```

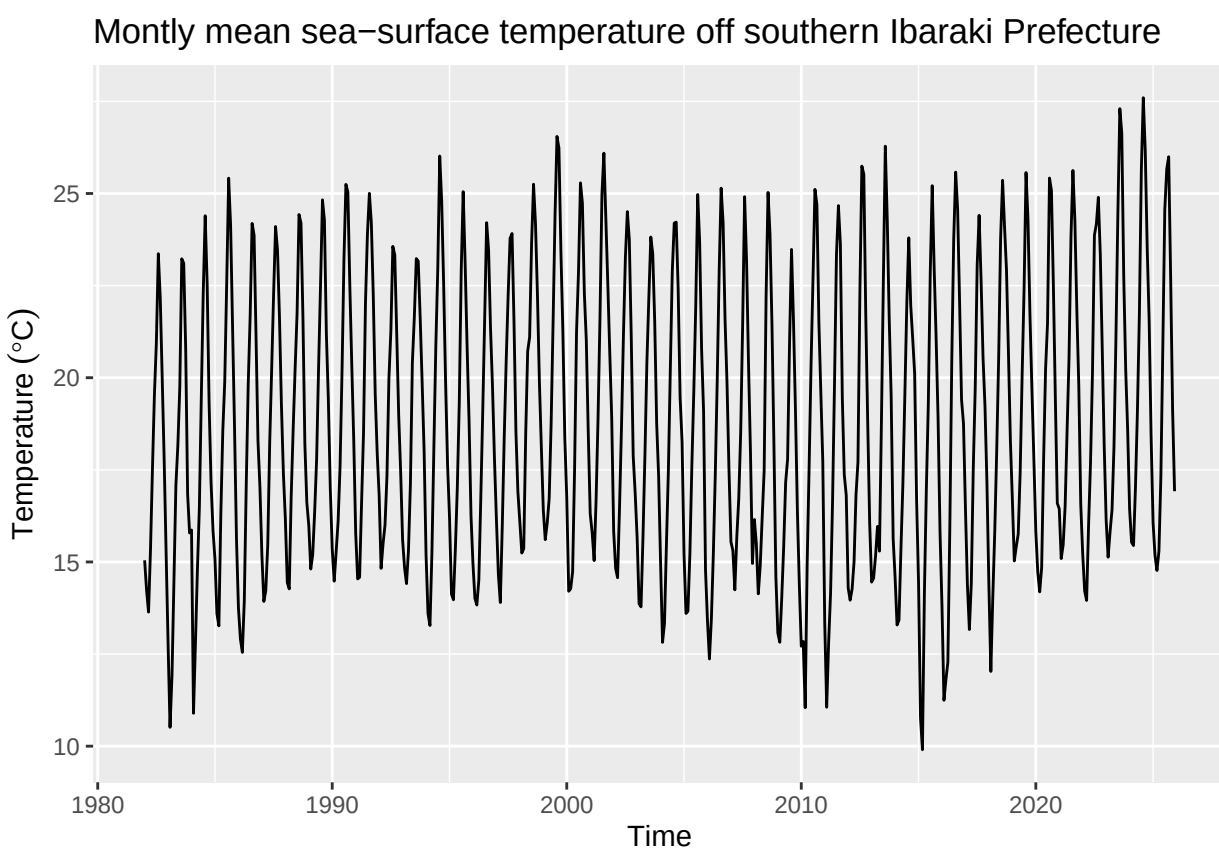
```
##           Jan      Feb      Mar      Apr      May      Jun      Jul      Aug
## 2001 16.30677 15.79500 15.03742 16.67533 18.87774 21.65100 24.97935 26.09419
##           Sep      Oct      Nov      Dec
## 2001 24.32400 22.53935 20.68533 18.92161
```

Plotting Monthly Time Series of Sea Surface Temperature

```r
monthly_ibaraki_sst_plot <- forecast::autoplot(monthly_ibaraki_sst) +
  labs(y = expression(Temperature~(degree*C)),
       x = "Time") +
  ggtitle("Montly mean sea-surface temperature off southern Ibaraki Prefecture")

#ggsave("monthly_SST_ibaraki_plot.png",
#       plot=monthly_ibaraki_sst_plot,
#       width=8,height=6)

plot(monthly_ibaraki_sst_plot)
```



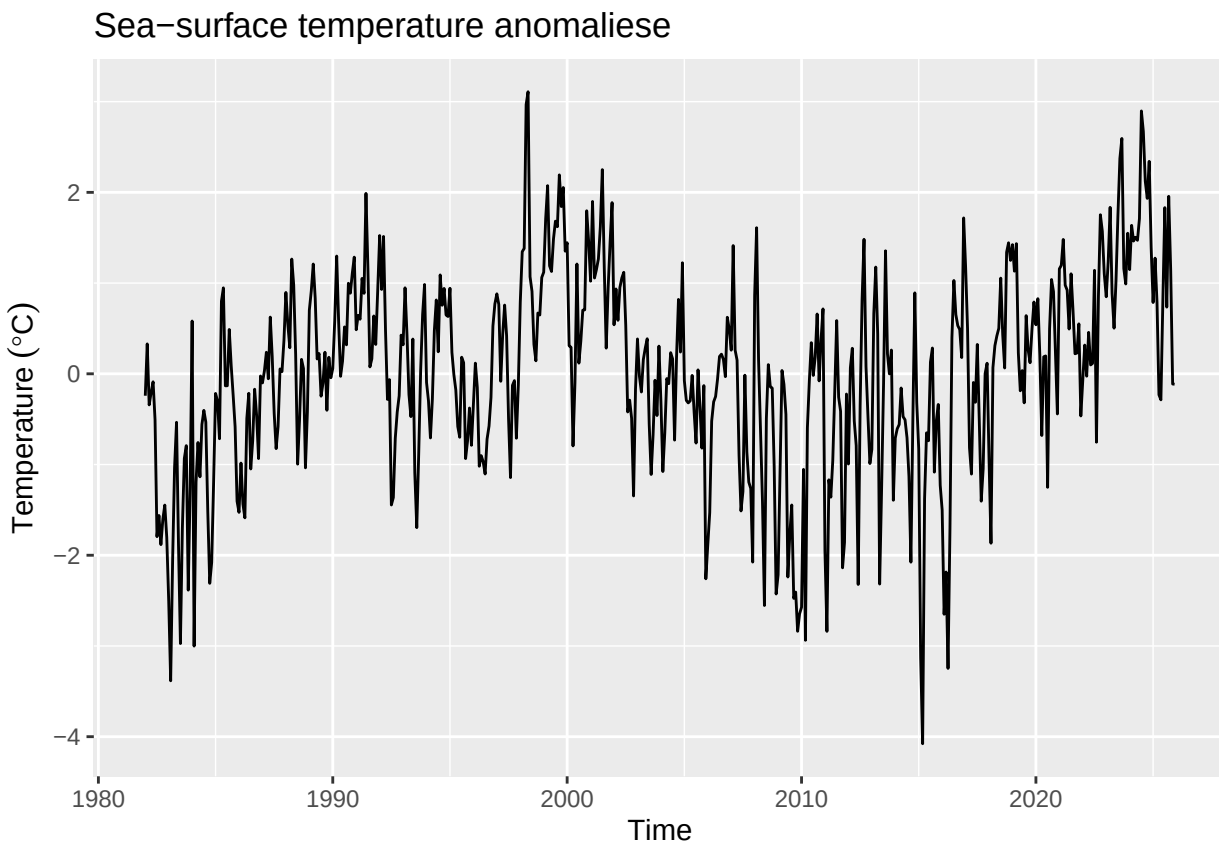
Plotting Time Series of Temperature Anomalies

```
# Generate temperature anomalies
monthly_ibaraki_sst_anomaly <- ThermoSSM::monthly_anomaly(monthly_ibaraki_sst)

monthly_ibaraki_sst_anomaly_plot <- forecast::autoplot(monthly_ibaraki_sst_anomaly) +
  labs(y = expression(Temperature~(degree*C)),
       x = "Time") +
  ggtitle("Sea-surface temperature anomaliese")

#ggsave("monthly_SST_dev_ibaraki.png",
#       plot=monthly_ibaraki_sst_anomaly_plot,
#       width=8,height=6)

plot(monthly_ibaraki_sst_anomaly_plot)
```



Plotting Monthly Seasonal Cycle

```
monthly_seasonal_cycle_SST_ibaraki <- ThermoSSM::mean_seasonal_cycle(monthly_ibaraki_sst)
summary(monthly_seasonal_cycle_SST_ibaraki)
```

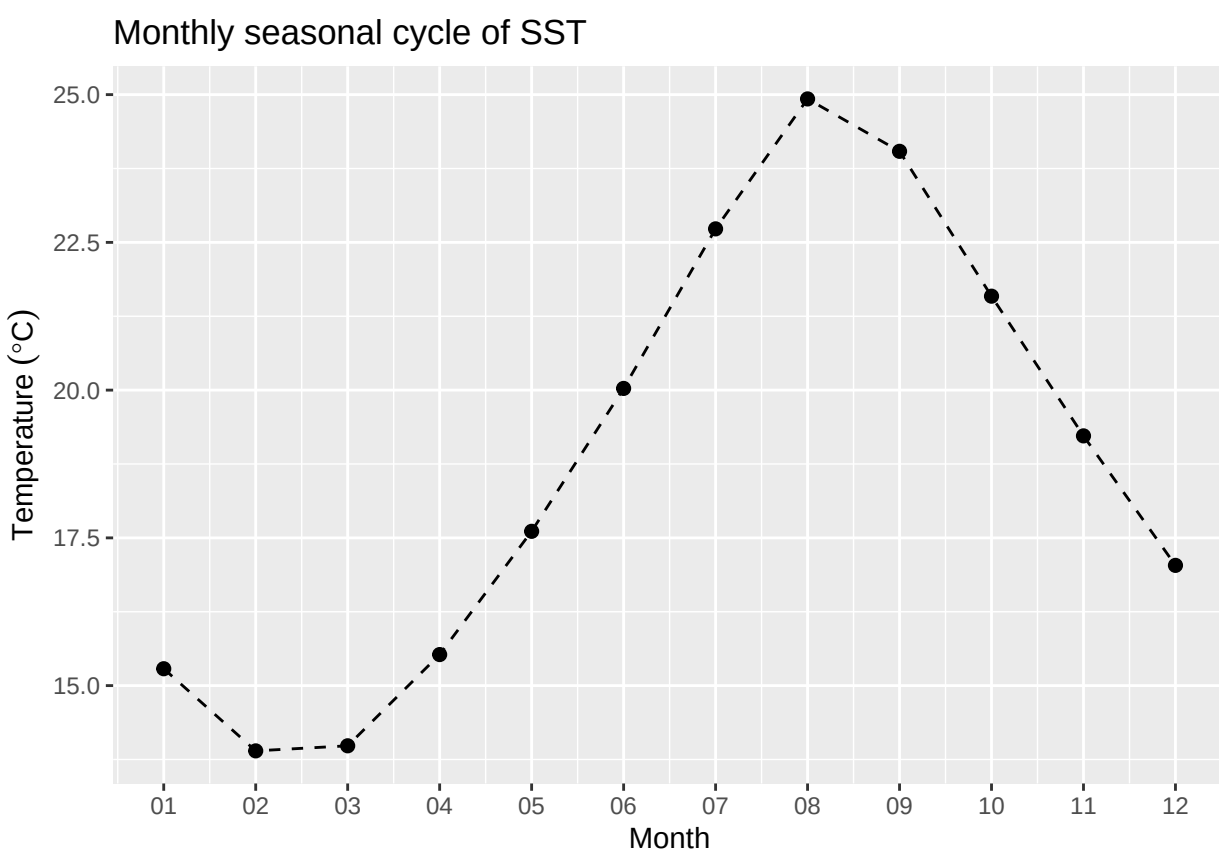
```
##      Month        Temperature
##  Min.   : 1.00   Min.   :13.90
##  1st Qu.: 3.75   1st Qu.:15.47
##  Median : 6.50   Median :18.42
##  Mean   : 6.50   Mean   :18.82
```

```
##  3rd Qu.: 9.25   3rd Qu.:21.87
##  Max.   :12.00   Max.   :24.93
```

```r
plt_monthly_seasonal_cycle_SST_ibaraki <- ggplot(
  data = monthly_seasonal_cycle_SST_ibaraki,
  aes(x = Month, y = Temperature)
) +
  geom_point(size = 2) +
  geom_line(linetype = "dashed") +
  labs(
    title = "Monthly seasonal cycle of SST",
    x = "Month",
    y = expression(Temperature~(degree*C))
  ) +
  scale_x_continuous(
    breaks = 1:12,
    labels = sprintf("%02d", 1:12)
  )


#ggsave("monthly_SST_seasonal_ibaraki.png",
#       plot=plt_monthly_seasonal_cycle_SST_ibaraki,
#       width=8,height=6)

plot(plt_monthly_seasonal_cycle_SST_ibaraki)
```



Applying a Linear Gaussian State-Space Model

```r
res <- lgssm(monthly_ibaraki_sst)

print(res)
```

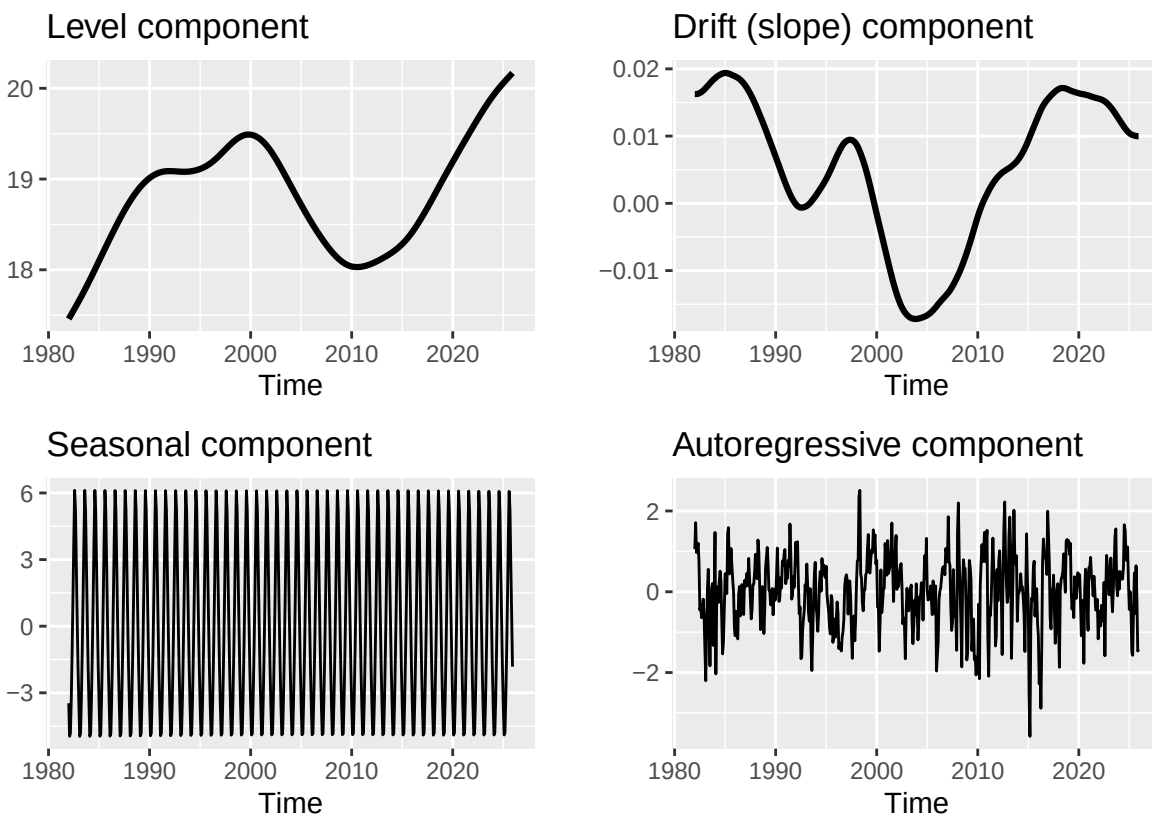
```
## ThermoSSM model fit
## ------------------
## Data:
##   Length      : 528
##   Frequency   : 12
##   Start / End : 1982-1  /  2025-12
##
## Optimization:
##   Converged   : TRUE
##   LogLik      : -603
##
## Use summary() for detailed results.
```

```r
summary(res)
```

```
## ThermoSSM summary
## -----------------
## Call:
## lgssm(ts_data = monthly_ibaraki_sst)
##
## Model fit:
##   Log-likelihood : -603
##   k              : 6
##   AIC            : 1218
##   Converged      : TRUE
##
## Variance parameters:
##   Observation (H): 1.72128e-43
##   State (Q trend): 5.008495e-06
##   State (Q season): 0.0001880851
##   State (Q ar): 0.5141963
## Coefficient of auto-regression parameters:
##   AR1: 0.6923188
##   AR2: -0.1245904
```

Plotting Level, Drift, Seasonal, and Auto-Regressive Components

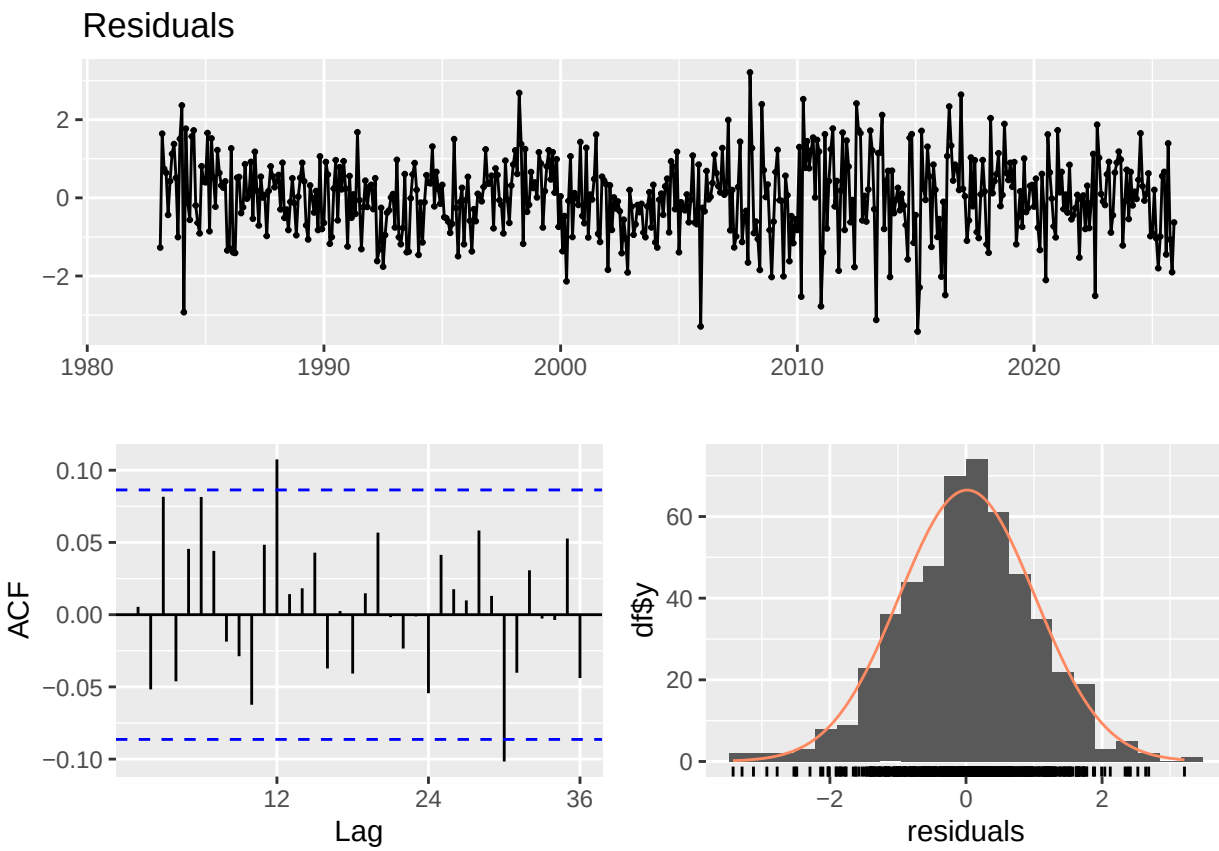
```r
#
plot(res)
```



Simple Model Diagnostics

```
## normality of residuals
#
std_obs_resid <- rstandard(res$kfs, type = "recursive")

# forecast   checkredisuals
# Ljung-Box : P > 0.05
checkresiduals(std_obs_resid)
```




```
##
##  Ljung-Box test
##
## data:  Residuals
## Q* = 28.235, df = 24, p-value = 0.2502
##
## Model df: 0.   Total lags used: 24
```

```r
#


## normality of residuals
# P > 0.05
shapiro.test(std_obs_resid)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  std_obs_resid
## W = 0.99552, p-value = 0.1467
```

Estimated Parameters and Components

```r
#
params <- ThermoSSM::extract_param(res)
params
```

```
##       Q_trend      Q_season           AR1           AR2          Q_ar
##  5.008495e-06  1.880851e-04  6.923188e-01 -1.245904e-01  5.141963e-01
##             H
##  1.721280e-43
```

```r
#
alpha_hat <- res$kfs$alphahat
head(alpha_hat)
```

```
##             level      slope sea_dummy1 sea_dummy2 sea_dummy3 sea_dummy4
## Jan 1982 17.45868 0.01625144  -3.470558  -1.801915  0.3388803  2.7790543
## Feb 1982 17.47493 0.01625139  -4.958038  -3.470558 -1.8019153  0.3388803
## Mar 1982 17.49118 0.01626273  -4.824835  -4.958038 -3.4705577 -1.8019153
## Apr 1982 17.50744 0.01628112  -3.279939  -4.824835 -4.9580380 -3.4705577
## May 1982 17.52372 0.01630892  -1.203128  -3.279939 -4.8248350 -4.9580380
## Jun 1982 17.54003 0.01635052   1.263210  -1.203128 -3.2799392 -4.8248350
##          sea_dummy5 sea_dummy6 sea_dummy7 sea_dummy8 sea_dummy9 sea_dummy10
## Jan 1982  5.2114469  6.1168103  3.8290115  1.2632100  -1.203128   -3.279939
## Feb 1982  2.7790543  5.2114469  6.1168103  3.8290115   1.263210   -1.203128
## Mar 1982  0.3388803  2.7790543  5.2114469  6.1168103   3.829011    1.263210
## Apr 1982 -1.8019153  0.3388803  2.7790543  5.2114469   6.116810    3.829011
## May 1982 -3.4705577 -1.8019153  0.3388803  2.7790543   5.211447    6.116810
## Jun 1982 -4.9580380 -3.4705577 -1.8019153  0.3388803   2.779054    5.211447
##          sea_dummy11    arima1      arima2
## Jan 1982   -4.824835 1.0560742 -0.06457844
## Feb 1982   -3.279939 1.7081096 -0.13157670
## Mar 1982   -1.203128 0.9726874 -0.21281405
## Apr 1982    1.263210 1.0918300 -0.12118751
## May 1982    3.829011 1.2019851 -0.13603153
## Jun 1982    6.116810 0.7197574 -0.14975580
```

```r
#
level_ts <- ThermoSSM::extract_level_ts(res)

#
drift_ts <- ThermoSSM::extract_drift_ts(res)

#
mean_drift_year <- mean(drift_ts) * 12
print(mean_drift_year)
```

```
## [1] 0.06181994
```

```r
# 1990
drift_1990s_per_year <- window(drift_ts,
                               start=c(1990,1),
                               end=c(1999,12)
                               ) %>%  mean()*12
print(drift_1990s_per_year)
```

```
## [1] 0.04776358
```

```r
# 2010
drift_2010s_per_year <- window(drift_ts,
                               start=c(2010,1),
                               end=c(2019,12)
                               ) %>%  mean()*12
```

```r
print(drift_2010s_per_year)
```

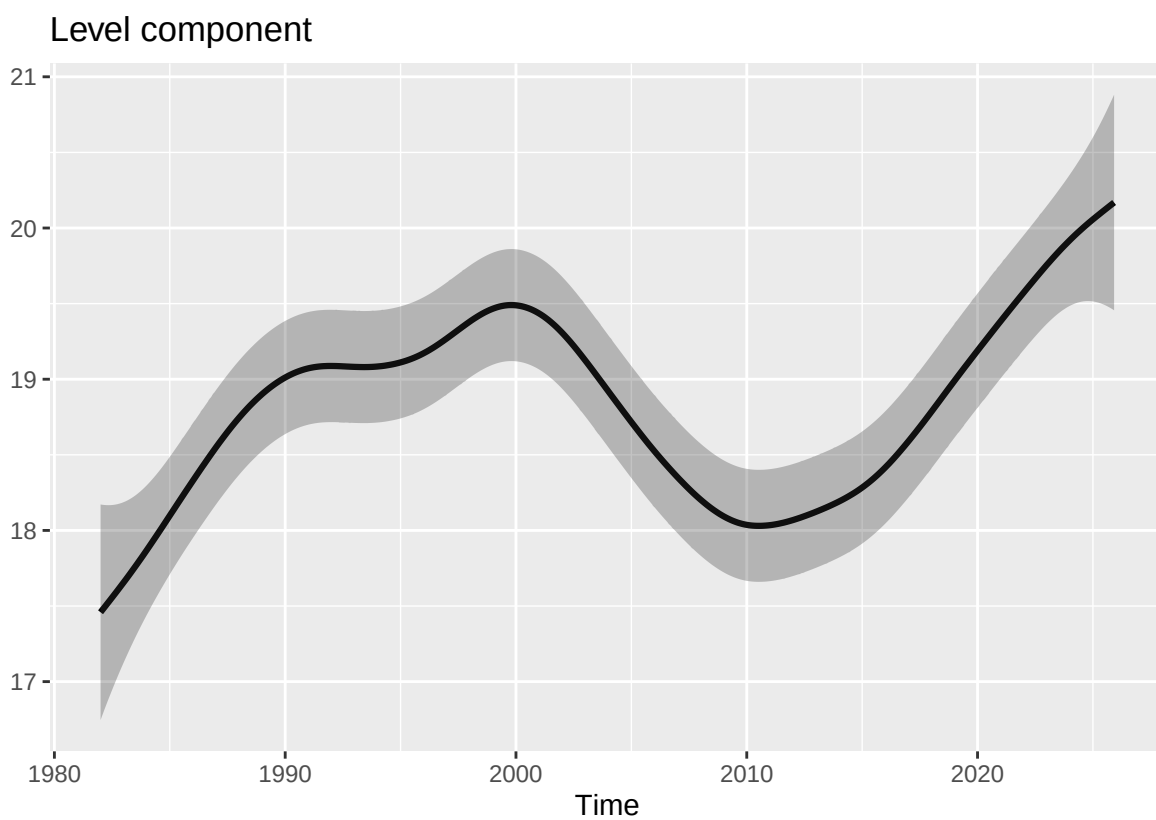
```
## [1] 0.1152325
```

Plotting Level Component with 95% Confidence Interval

```r
plt_level_ci <- plot(res,
                     components = "level",
                     ci = TRUE,
                     ci_level = 0.95
                     )

#ggsave("level_plot.png",
#       width=6, height=4,
#       plot = plt_level_ci)

plot(plt_level_ci)
```

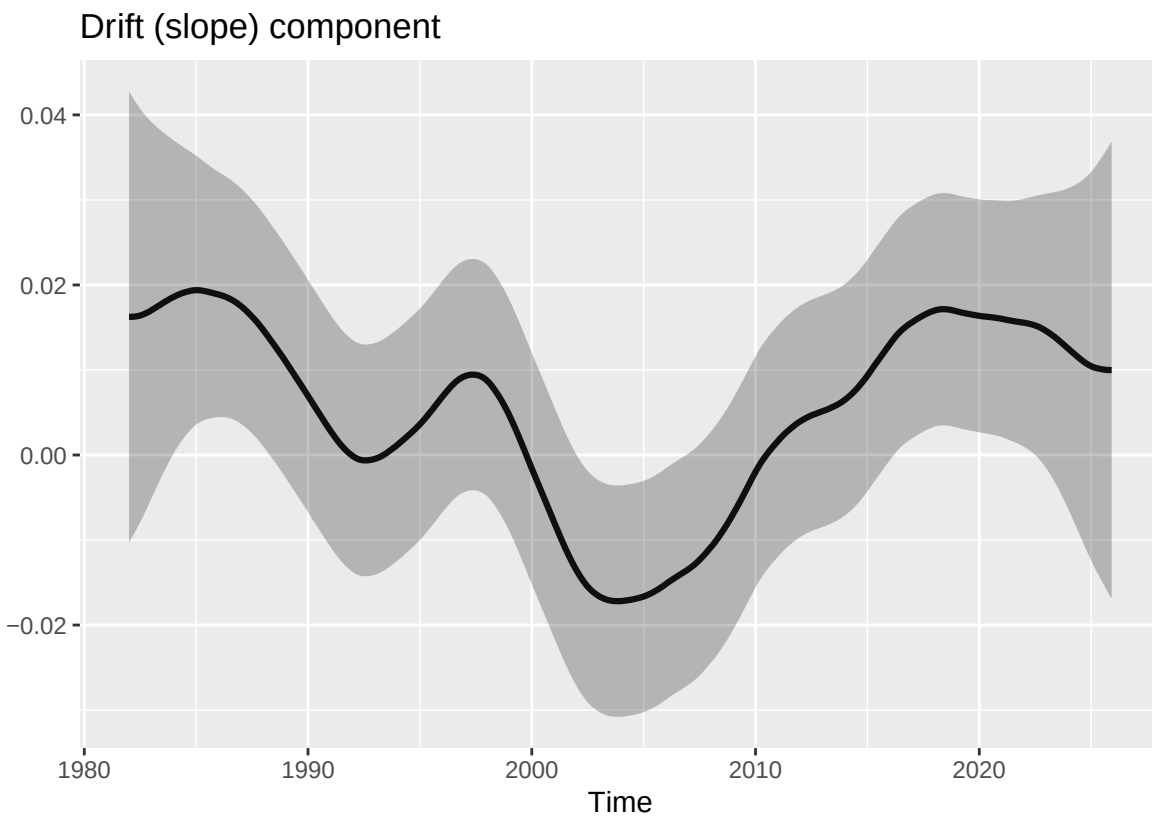


Plotting Drift Component with 95% Confidence Interval

```r
plt_drift_ci <- plot(res,
                     components = "drift",
                     ci = TRUE,
                     ci_level = 0.95
                     )
```

```r
#ggsave("drift_plot.png",
#       width=6, height=4,
#       plot = plt_drift_ci)

plot(plt_drift_ci)
```



Appendix: Handling Raw Temperature Data

This appendix describes how the original temperature time series data are prepared prior to the state-space analysis presented in the main text. We outline the procedures for handling raw observations.

```r
##========================
## Example 1
# Monthly temperature data
original_data <- data.frame(
  Year=c(2000,2000,2000,2000,2000,2000,2000,2000,2000,2000,2000,2000),
  Month=c(1,2,3,4,5,6,7,8,9,10,11,12),
  Temp=   c(5.2, 6.1, 9.3, 13.5, 17.8, 21.0,24.3, 25.1, 22.0, 17.1, 11.2,7.0))

head(original_data)
```

```
##   Year Month Temp
## 1 2000     1  5.2
## 2 2000     2  6.1
## 3 2000     3  9.3
```

```
## 4 2000     4 13.5
## 5 2000     5 17.8
## 6 2000     6 21.0
```

```r
# Convert ts object
original_ts <- ts(
  matrix(original_data$Temp),
  start = c(original_data$Year[1],original_data$Month[1]),
  frequency = 12
)

# Important!!: Exe colnames() to add "Temp" name to ts object for the model
colnames(original_ts) <- c("Temp")

frequency(original_ts)
```

```
## [1] 12
```

```r
start(original_ts)
```

```
## [1] 2000    1
```

```r
end(original_ts)
```

```
## [1] 2000   12
```

```r
time(original_ts)
```

```
##           Jan      Feb      Mar      Apr      May      Jun      Jul      Aug
## 2000 2000.000 2000.083 2000.167 2000.250 2000.333 2000.417 2000.500 2000.583
##           Sep      Oct      Nov      Dec
## 2000 2000.667 2000.750 2000.833 2000.917
```

```r
original_ts
```

```
##       Jan  Feb  Mar  Apr  May  Jun  Jul  Aug  Sep  Oct  Nov  Dec
## 2000  5.2  6.1  9.3 13.5 17.8 21.0 24.3 25.1 22.0 17.1 11.2  7.0
```

```r
##=========================
## Example 2
# loading and checking raw csv data
input_csv <- "./data/test_data.csv"
raw_csv <- read_csv(input_csv)
head(raw_csv)
```

```
## # A tibble: 6 x 3
##    Year Month  Temp
##   <dbl> <dbl> <dbl>
## 1  2002     1 -15.1
## 2  2002     2 -13.3
## 3  2002     3 -10.6
## 4  2002     4  -0.8
## 5  2002     5   3.5
## 6  2002     6   6.2
```

```r
# Load original csv file and then convert to monthly ts object
# help(ThermoSSM::monthly_csv2ts)
test_ts <- ThermoSSM::monthly_csv2ts(input_csv)
head(test_ts)
```

```
##        Jan   Feb   Mar   Apr   May   Jun
## 2002 -15.1 -13.3 -10.6  -0.8   3.5   6.2
```

```r
frequency(test_ts)
```

```
## [1] 12
```

```r
start(test_ts)
```

```
## [1] 2002    1
```

```r
end(test_ts)
```

```
## [1] 2025   12
```

```r
cycle(test_ts) %>% head()
```

```
##      Jan Feb Mar Apr May Jun
## 2002   1   2   3   4   5   6
```

```r
time(test_ts) %>% head()
```

```
##           Jan      Feb      Mar      Apr      May      Jun
## 2002 2002.000 2002.083 2002.167 2002.250 2002.333 2002.417
```